

Takayasu Sakurai

其他

基本信息

姓名: Takayasu Sakurai

所属院校: 东京大学

地区: Japan

学科领域

微电子科学与工程

集成电路设计与集成系统

电气工程及其自动化

大功率半导体科学与工程

柔性电子学

通信工程

研究领域

低功耗电路与能量管理

电力电子与电机驱动

数字集成电路设计

智能功率集成电路

射频集成电路

柔性器件制备与工艺

超大规模集成电路设计

无线通信与智能天线

模拟与数模混合电路

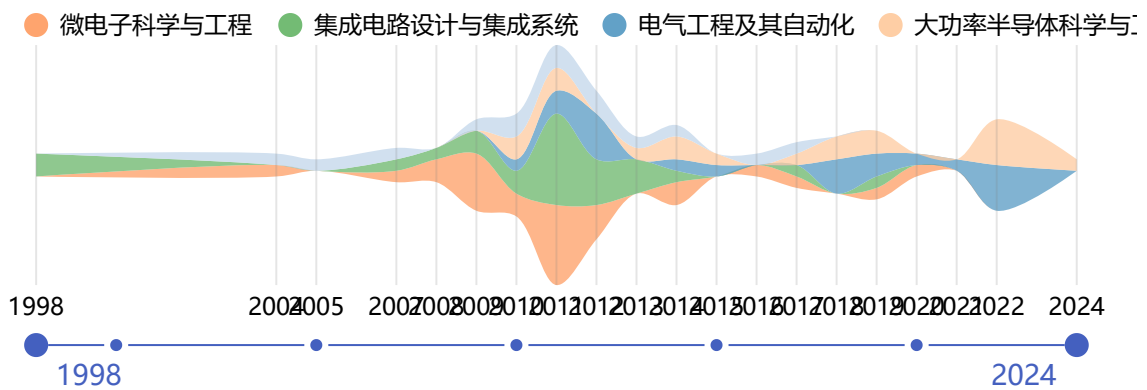
微电子器件与制造

+3

个人简介

Takayasu Sakurai is a distinguished professor at the University of Tokyo, specializing in energy-efficient micro-electronics design, power electronics, and flexible electronics. With a verified email at iis.u-tokyo.ac.jp, Sakurai has made significant contributions to the field through numerous publications, including pioneering work on flexible pressure sensor matrices for artificial skin applications and the development of organic nonvolatile memory transistors for flexible sensor arrays. His research has been widely recognized, with several papers garnering over 1000 citations each.

近年发表的论文



Estimating of IGBT Bond Wire Lift-Off Trend Using Convolutional Neural Network (CNN)

IEEE ACCESS

Mamee, Thatree; Lou, Zaiqi; Hata, Katsuhiro; Takamiya, Makoto; Sakurai, Takayasu; Nishizawa, Shin-Ichi; Saito, Wataru

被引用次数: 0

2024

A 6.78-MHz Multiple-Transmitter Wireless Power Transfer System With Efficiency Maximization by Adaptive Magnetic Field Adder IC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Qiu, Hao; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 7

2022

Equalization of DC and Surge Components of Drain Current of Two Parallel-Connected SiC MOSFETs Using Single-Input Dual-Output Digital Gate Driver IC

2022 IEEE APPLIED POWER ELECTRONICS CONFERENCE AND EXPOSITION, APEC

Horii, Kohei; Morikawa, Ryuzo; Katada, Ryunosuke; Hata, Katsuhiro; Sakurai, Takayasu; Hayashi, Shin-Ichiro; Wada, Keiji; Omura, Ichiro; Takamiya, Makoto

被引用次数: 5

2022

An Optimization Method of a Digital Active Gate Driver Under Continuous Switching Operation Being Capable of Suppressing Surge Voltage and Power Loss in PWM Inverters

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Yamaguchi, Daiki; Cheng, Yu Shan; Mannen, Tomoyuki; Obara, Hidemine; Wada, Keiji; Sai, Toru; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 8

2022

High-Speed Searching of Optimum Switching Pattern for Digital Active Gate Drive to Adapt to Various Load Conditions

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Cheng, Yu Shan; Yamaguchi, Daiki; Mannen, Tomoyuki; Wada, Keiji; Sai, Toru; Miyazaki, Koutarou; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 9

2022

Analysis and Mitigation of Coupling-Dependent Data Flipping in Wireless Power and Data Transfer System

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Qiu, Hao; Jiang, Yuntao; Shi, Yi; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 3

2021

Trillion-node engine: open-innovation IoT/CPS platform-pioneering future of IoT/CPS for everyone, by everyone

JAPANESE JOURNAL OF APPLIED PHYSICS

Sakurai, Takayasu

被引用次数: 3

2020

Digital Transmitter Coil for Wireless Power Transfer Robust Against Variation of Distance and Lateral Misalignment

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Qiu, Hao; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 23

2020

Optimization Platform to Find a Switching Pattern of Digital Active Gate Drive for Reducing Both Switching Loss and Surge Voltage

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Cheng, Yu Shan; Mannen, Tomoyuki; Wada, Keiji; Miyazaki, Koutarou; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 27

2019

An 11-nW CMOS Temperature-to-Digital Converter Utilizing Sub-Threshold Current at Sub-Thermal Drain Voltage

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Someya, Teruki; Islam, A. K. M. Mahfuzul; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 60

2019

A 0.90-4.39-V Detection Voltage Range, 56-Level Programmable Voltage Detector Using Fine Voltage-Step Subtraction for Battery Management

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Someya, Teruki; Matsunaga, Kenichi; Morimura, Hiroki; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 10

2019

Clocked Hysteresis Control Scheme With Power-Law Frequency Scaling in Buck Converter to Improve Light-Load Efficiency for IoT Sensor Nodes

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wu, Chung-Shiang; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 13

2018

Active Gate Control in Half-Bridge Inverters Using Programmable Gate Driver ICs to Improve Both Surge Voltage and Converter

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Obara, Hidemine; Wada, Keiji; Miyazaki, Koutarou; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 57

2018

2/3 and 1/2 Reconfigurable Switched Capacitor DC-DC Converter With 92.9% Efficiency at 62 mW/mm² Using Driver Amplitude Doubler

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sai, Toru; Yamauchi, Yoshitaka; Kando, Hajime; Funaki, Tatsuya; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 12

2018

A Low-Power CMOS Crystal Oscillator Using a Stacked-Amplifier Architecture

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Iguchi, Shunta; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 24

2017

Design and Analysis of Ultra-Low Power Glitch-Free Programmable Voltage Detector Based on Multiple Voltage Copier

IEICE TRANSACTIONS ON ELECTRONICS

Someya, Teruki; Fuketa, Hiroshi; Matsunaga, Kenichi; Morimura, Hiroki; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 0

2017

General-Purpose Clocked Gate Driver IC With Programmable 63-Level Drivability to Optimize Overshoot and Energy Loss in Switching by a Simulated Annealing Algorithm

IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS

Miyazaki, Koutarou; Abe, Seiya; Tsukuda, Masanori; Omura, Ichiro; Wada, Keiji; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 93

2017

Programmable Neuron Array Based on a 2-Transistor Multiplier Using Organic Floating-Gate for Intelligent Sensors

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Islam, A. K. M. Mahfuzul; Hamamatsu, Masamune; Yokota, Tomoyuki; Lee, Sunghoon; Yukita, Wakako; Takamiya, Makoto; Someya, Takao; Sakurai, Takayasu

被引用次数: 12

2017

Variation-Tolerant Quick-Start-Up CMOS Crystal Oscillator With Chirp Injection and Negative Resistance Booster

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Iguchi, Shunta; Fuketa, Hiroshi; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 38

2016

High-Frequency, Conformable Organic Amplifiers

ADVANCED MATERIALS

Reuveny, Amir; Lee, Sunghoon; Yokota, Tomoyuki; Fuketa, Hiroshi; Siket, Christian M.; Lee, Sungwon; Sekitani, Tsuyoshi; Sakurai, Takayasu; Bauer, Siegfried; Someya, Takao

被引用次数: 50

2016

Wireless Power Transfer With Zero-Phase-Difference Capacitance Control

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Iguchi, Shunta; Yeon, Pyungwoo; Fuketa, Hiroshi; Ishida, Koichi; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 22

2015

Design guidelines to achieve minimum energy operation for ultra low voltage tunneling FET logic circuits

JAPANESE JOURNAL OF APPLIED PHYSICS

Fuketa, Hiroshi; Yoshioka, Kazuaki; Fukuda, Koichi; Mori, Takahiro; Ota, Hiroyuki; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 13

2015

Design Method of Class-F Power Amplifier With Output Power of -20 dBm and Efficient Dual Supply Voltage Transmitter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Iguchi, Shunta; Saito, Akira; Watanabe, Kazunori; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 17

2014

1 μ m-Thickness Ultra-Flexible and High Electrode-Density Surface Electromyogram Measurement Sheet With 2 V Organic Transistors for Prosthetic Hand Control

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Fuketa, Hiroshi; Yoshioka, Kazuaki; Shinozuka, Yasuhiro; Ishida, Koichi; Yokota, Tomoyuki; Matsuhisa, Naoji; Inoue, Yusuke; Sekino, Masaki; Sekitani, Tsuyoshi; Takamiya, Makoto; Someya, Takao; Sakurai, Takayasu

被引用次数: 59

2014

A 0.6 V Input CCM/DCM Operating Digital Buck Converter in 40 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Zhang, Xin; Chen, Po-Hung; Okuma, Yasuyuki; Ishida, Koichi; Ryu, Yoshikatsu; Watanabe, Kazunori; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 61

2014

Intermittent Resonant Clocking Enabling Power Reduction at Any Clock Frequency for Near/Sub-Threshold Logic Circuits

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fuketa, Hiroshi; Nomura, Masahiro; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 9

2014

Analog-assisted digital low dropout regulator with fast transient response and low output ripple

JAPANESE JOURNAL OF APPLIED PHYSICS

Mori, Kazuaki; Okuma, Yasuyuki; Zhang, Xin; Fuketa, Hiroshi; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 3

2014

Increase of Crosstalk Noise Due to Imbalanced Threshold Voltage Between nMOS and pMOS in Subthreshold Logic Circuits

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Fuketa, Hiroshi; Takahashi, Ryo; Takamiya, Makoto; Nomura, Masahiro; Shinohara, Hirofumi; Sakurai, Takayasu

Highly Energy-Efficient SRAM With Hierarchical Bit Line Charge-Sharing Method Using Non-Selected Bit Line Charges

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Miyano, Shinji; Moriwaki, Shinichi; Yamamoto, Yasue; Kawasumi, Atsushi; Suzuki, Toshikazu; Sakurai, Takayasu; Shinohara, Hirofumi

被引用次数: 13

2013

Insole Pedometer With Piezoelectric Energy Harvester and 2 V Organic Circuits

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ishida, Koichi; Huang, Tsung-Ching; Honda, Kentaro; Shinozuka, Yasuhiro; Fuketa, Hiroshi; Yokota, Tomoyuki; Zschieschang, Ute; Klauk, Hagen; Tortissier, Gregory; Sekitani, Tsuyoshi; Toshiyoshi, Hiroshi; Takamiya, Makoto; Someya, Takao; Sakurai, Takayasu

被引用次数: 82

2013

Minimizing Energy of Integer Unit by Higher Voltage Flip-Flop: VDDmin-Aware Dual Supply Voltage Technique

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Fuketa, Hiroshi; Hirairi, Koji; Yasufuku, Tadashi; Takamiya, Makoto; Nomura, Masahiro; Shinohara, Hirofumi; Sakurai, Takayasu

被引用次数: 3

2013

On-Chip Measurement System for Within-Die Delay Variation of Individual Standard Cells in 65-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Xin; Ishida, Koichi; Fuketa, Hiroshi; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 6

2012

Large Within-Die Gate Delay Variations in Sub-Threshold Logic Circuits at Low Temperature

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Takahashi, Ryo; Takata, Hidehiro; Yasufuku, Tadashi; Fuketa, Hiroshi; Takamiya, Makoto; Nomura, Masahiro; Shinohara, Hirofumi; Sakurai, Takayasu

被引用次数: 9

2012

A 1-V-Input Switched-Capacitor Voltage Converter With Voltage-Reference-Free Pulse-Density Modulation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xin; Pu, Yu; Ishida, Koichi; Ryu, Yoshikatsu; Okuma, Yasuyuki; Chen, Po-Hung; Watanabe, Kazunori; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 10

2012

An 80 mV Startup Dual-Mode Boost Converter by Charge-Pumped Pulse Generator and Threshold Voltage Tuned Oscillator With Hot Carrier Injection

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Po-Hung; Zhang, Xin; Ishida, Koichi; Okuma, Yasuyuki; Ryu, Yoshikatsu; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 48

2012

Sheet-Type Flexible Organic Active Matrix Amplifier System Using Pseudo-CMOS Circuits With Floating-Gate Structure

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yokota, Tomoyuki; Sekitani, Tsuyoshi; Tokuhara, Takeyoshi; Take, Naoya; Zschieschang, Ute; Klauk, Hagen; Takimiya, Kazuo; Huang, Tsung-Ching; Takamiya, Makoto; Sakurai, Takayasu; Someya, Takao

被引用次数: 90

2012

Startup Techniques for 95 mV Step-Up Converter by Capacitor Pass-On Scheme and VTH-Tuned Oscillator With Fixed Charge Programming

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Po-Hung; Ishida, Koichi; Ikeuchi, Katsuyuki; Zhang, Xin; Honda, Kentaro; Okuma, Yasuyuki; Ryu, Yoshikatsu; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 119

2012

A 315 MHz Power-Gated Ultra Low Power Transceiver in 40 nm CMOS for Wireless Sensor Network

IEICE TRANSACTIONS ON ELECTRONICS

Liu, Lechang; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 2

2012

EMI Camera LSI (EMcam) with On-Chip Loop Antenna Matrix to Measure EMI Noise Spectrum and Distribution

IEICE TRANSACTIONS ON ELECTRONICS

Masunaga, Naoki; Ishida, Koichi; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 0

2012

Positioning-Free Resonant Wireless Power Transmission Sheet With Staggered Repeater Coil Array (SRCA)

IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS

Mori, Kazuaki; Lim, Hyunkeun; Iguchi, Shunta; Ishida, Koichi; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 36

2012

A 100-V AC Energy Meter Integrating 20-V Organic CMOS Digital and Analog Circuits With a Floating Gate for Process Variation Compensation and a 100-V Organic pMOS Rectifier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ishida, Koichi; Huang, Tsung-Ching; Honda, Kentaro; Sekitani, Tsuyoshi; Nakajima, Hiroyoshi; Maeda, Hiroki; Takamiya, Makoto; Someya, Takao; Sakurai, Takayasu

被引用次数: 20

2012

Control of threshold voltage in low-voltage organic complementary inverter circuits with floating gate structures

APPLIED PHYSICS LETTERS

Yokota, Tomoyuki; Nakagawa, Takashi; Sekitani, Tsuyoshi; Noguchi, Yoshiaki; Fukuda, Kenjiro;
Zschieschang, Ute; Klauk, Hagen; Takeuchi, Ken; Takamiya, Makoto; Sakurai, Takayasu; Someya, Takao
被引用次数: 52 2011

0.18-V Input Charge Pump with Forward Body Bias to Startup Boost Converter for Energy Harvesting Applications

IEICE TRANSACTIONS ON ELECTRONICS

Chen, Po-Hung; Ishida, Koichi; Zhang, Xin; Okuma, Yasuyuki; Ryu, Yoshikatsu; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 7 2011

Power Supply Voltage Dependence of Within-Die Delay Variation of Regular Manual Layout and Irregular Place-and-Route Layout

IEICE TRANSACTIONS ON ELECTRONICS

Yasufuku, Tadashi; Nakamura, Yasumi; Piao, Zhe; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 0 2011

1 Gb/s, 50 μm x 50 μm Pads on Board Wireless Connector Based on Track-and-Charge Scheme Allowing Contacted Signaling

IEICE TRANSACTIONS ON ELECTRONICS

Ikeuchi, Katsuyuki; Kusamitsu, Hideki; Daito, Mutsuo; Kim, Gil-Su; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 1 2011

Through-Silicon Via Design for a 3-D Solid-State Drive System with Boost Converter in a Package

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Johguchi, Koh; Hatanaka, Teruyoshi; Ishida, Koichi; Yasufuku, Tadashi; Takamiya, Makoto; Sakurai, Takayasu; Takeuchi, Ken

被引用次数: 9 2011

0.5-V Input Digital Low-Dropout Regulator (LDO) with 98.7% Current Efficiency in 65 nm CMOS

IEICE TRANSACTIONS ON ELECTRONICS

Okuma, Yasuyuki; Ishida, Koichi; Ryu, Yoshikatsu; Zhang, Xin; Chen, Po-Hung; Watanabe, Kazunori; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 14 2011

A Variable Output Voltage Switched-Capacitor DC-DC Converter with Pulse Density and Width Modulation (PDWM) for 57% Ripple Reduction at Low Output Voltage

IEICE TRANSACTIONS ON ELECTRONICS

Zhang, Xin; Pu, Yu; Ishida, Koichi; Ryu, Yoshikatsu; Okuma, Yasuyuki; Chen, Po-Hung; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 2 2011

0.6 V Voltage Shifter and Clocked Comparator for Sampling Correlation-Based Impulse Radio UWB Receiver

IEICE TRANSACTIONS ON ELECTRONICS

Liu, Lechang; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 0

2011

Organic Pseudo-CMOS Circuits for Low-Voltage Large-Gain High-Speed Operation

IEEE ELECTRON DEVICE LETTERS

Fukuda, Kenjiro; Sekitani, Tsuyoshi; Yokota, Tomoyuki; Kuribara, Kazunori; Huang, Tsung-Ching; Sakurai, Takayasu; Zschieschang, Ute; Klauk, Hagen; Ikeda, Masaaki; Kuwabara, Hirokazu; Yamamoto, Tatsuya; Takimiya, Kazuo; Cheng, Kwang-Ting; Someya, Takao

被引用次数: 54

2011

Post-Silicon Clock Deskew Employing Hot-Carrier Injection Trimming With On-Chip Skew Monitoring and Auto-Stressing Scheme for Sub/Near Threshold Digital Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Pu, Yu; Zhang, Xin; Ikeuchi, Katsuyuki; Muramatsu, Atsushi; Kawasumi, Atsushi; Takamiya, Makoto; Nomura, Masahiro; Shinohara, Hirofumi; Sakurai, Takayasu

被引用次数: 2

2011

Capacitively Coupled Non-Contact Probing Circuits for Membrane-Based Wafer-Level Simultaneous Testing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Daito, Mutsuo; Nakata, Yoshiro; Sasaki, Satoshi; Gomyo, Hiroyuki; Kusamitsu, Hideki; Komoto, Yoshio; Iizuka, Kunihiro; Ikeuchi, Katsuyuki; Kim, Gil Su; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 13

2011

A Charge-Domain Auto- and Cross-Correlation Based Data Synchronization Scheme With Power- and Area-Efficient PLL for Impulse Radio UWB Receiver

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Liu, Lechang; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 14

2011

1.8 V Low-Transient-Energy Adaptive Program-Voltage Generator Based on Boost Converter for 3D-Integrated NAND Flash SSD

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ishida, Koichi; Yasufuku, Tadashi; Miyamoto, Shinji; Nakai, Hiroto; Takamiya, Makoto; Sakurai, Takayasu; Takeuchi, Ken

被引用次数: 27

2011

User Customizable Logic Paper (UCLP) With Sea-Of Transmission-Gates (SOTG) of 2-V Organic CMOS and Ink-Jet Printed Interconnects

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ishida, Koichi; Masunaga, Naoki; Takahashi, Ryo; Sekitani, Tsuyoshi; Shino, Shigeki; Zschieschang, Ute; Klauk, Hagen; Takamiya, Makoto; Someya, Takao; Sakurai, Takayasu

被引用次数: 29

2011

A 4 V Operation, Flexible Braille Display Using Organic Transistors, Carbon Nanotube Actuators, and Organic Static Random-Access Memory

ADVANCED FUNCTIONAL MATERIALS

Fukuda, Kenjiro; Sekitani, Tsuyoshi; Zschieschang, Ute; Klauk, Hagen; Kuribara, Kazunori; Yokota, Tomoyuki; Sugino, Takushi; Asaka, Kinji; Ikeda, Masaaki; Kuwabara, Hirokazu; Yamamoto, Tatsuya; Takimiya, Kazuo; Fukushima, Takanori; Aida, Takuzo; Takamiya, Makoto; Sakurai, Takayasu; Someya, Takao
被引用次数: 122 2011

Power Gating: Circuits, Design Methodologies, and Best Practice for Standard-Cell VLSI Designs

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Shin, Youngsoo; Seomun, Jun; Choi, Kyu-Myung; Sakurai, Takayasu

被引用次数: 72 2010

2 Gb/s 15 pJ/b/chip Inductive-Coupling Programmable Bus for NAND Flash Memory Stacking

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Saito, Mitsuko; Sugimori, Yasufumi; Kohama, Yoshinori; Yoshida, Yoichi; Miura, Noriyuki; Ishikuro, Hiroki; Sakurai, Takayasu; Kuroda, Tadahiyo

被引用次数: 8 2010

Stretchable EMI Measurement Sheet With 8 x 8 Coil Array, 2 V Organic CMOS Decoder, and 0.18 μm Silicon CMOS LSIs for Electric and Magnetic Field Detection

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ishida, Koichi; Masunaga, Naoki; Zhou, Zhiwei; Yasufuku, Tadashi; Sekitani, Tsuyoshi; Zschieschang, Ute; Klauk, Hagen; Takamiya, Makoto; Someya, Takao; Sakurai, Takayasu

被引用次数: 22 2010

A 1.76 mW, 100 Mbps Impulse Radio UWB Receiver with Multiple Sampling Correlators Eliminating Need for Phase Synchronization in 65-nm CMOS

IEICE TRANSACTIONS ON ELECTRONICS

Liu, Lechang; Zhou, Zhiwei; Sakurai, Takayasu; Takamiya, Makoto

被引用次数: 0 2010

Large-Area Flexible Ultrasonic Imaging System With an Organic Transistor Active Matrix

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kato, Yusaku; Sekitani, Tsuyoshi; Noguchi, Yoshiaki; Yokota, Tomoyuki; Takamiya, Makoto; Sakurai, Takayasu; Someya, Takao

被引用次数: 81 2010

Inductor and TSV Design of 20-V Boost Converter for Low Power 3D Solid State Drive with NAND Flash Memories

IEICE TRANSACTIONS ON ELECTRONICS

Yasufuku, Tadashi; Ishida, Koichi; Miyamoto, Shinji; Nakai, Hiroto; Takamiya, Makoto; Sakurai, Takayasu; Takeuchi, Ken

被引用次数: 5 2010

Difficulty of Power Supply Voltage Scaling in Large Scale Subthreshold Logic Circuits

IEICE TRANSACTIONS ON ELECTRONICS

Yasufuku, Tadashi; Niiyama, Taro; Piao, Zhe; Ishida, Koichi; Murakata, Masami; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 6

2010

A 100 Mbps, 4.1 pJ/bit Threshold Detection-Based Impulse Radio UWB Transceiver in 90 nm CMOS

IEICE TRANSACTIONS ON ELECTRONICS

Liu, Lechang; Miyamoto, Yoshio; Zhou, Zhiwei; Sakaida, Kosuke; Ryu, Jisun; Ishida, Koichi; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 6

2009

An On-Chip Noise Canceller with High Voltage Supply Lines for Nanosecond-Range Power Supply Noise

IEICE TRANSACTIONS ON ELECTRONICS

Nakamura, Yasumi; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 1

2009

Organic Nonvolatile Memory Transistors for Flexible Sensor Arrays

SCIENCE

Sekitani, Tsuyoshi; Yokota, Tomoyuki; Zschieschang, Ute; Klauk, Hagen; Bauer, Siegfried; Takeuchi, Ken; Takamiya, Makoto; Sakurai, Takayasu; Someya, Takao

被引用次数: 943

2009

Velocity Saturation Effects in a Short Channel Si-MOSFET and its Small Signal Characteristics

JOURNAL OF THE KOREAN PHYSICAL SOCIETY

Hwang, Sanghoon; Im, Hyunsik; Song, Minkyu; Ishida, Koichi; Hiramoto, Toshiro; Sakurai, Takayasu

被引用次数: 3

2009

A High-Speed Inductive-Coupling Link With Burst Transmission

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Miura, Noriyuki; Kohama, Yoshinori; Sugimori, Yasfumi; Ishikuro, Hiroki; Sakurai, Takayasu; Kuroda, Tadahiro

被引用次数: 51

2009

Printed Nonvolatile Memory for a Sheet-Type Communication System

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sekitani, Tsuyoshi; Zaitzu, Koichiro; Noguchi, Yoshiaki; Ishibe, Kiyoshiro; Takamiya, Makoto; Sakurai, Takayasu; Someya, Takao

被引用次数: 125

2009

A 107-pJ/bit 100-kb/s 0.18- μ m Capacitive-Coupling Transceiver With Data Edge Signaling and DC Power-Free Pulse Detector for Printable Communication Sheet

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Liu, Lechang; Takamiya, Makoto; Sekitani, Tsuyoshi; Noguchi, Yoshiaki; Nakano, Shintaro; Zaitzu, Koichiro; Kuroda, Tadahiro; Someya, Takao; Sakurai, Takayasu

被引用次数: 3

2009

A 25-mV-Sensitivity 2-Gb/s Optimum-Logic-Threshold Capacitive-Coupling Receiver for Wireless Wafer Probing Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Gil-Su; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 182

2009

Backgate Bias Accelerator for sub-100 ns Sleep-to-Active Modes Transition Time

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Levacq, David; Takamiya, Makoto; Sakurai, Takayasu

被引用次数: 7

2008

A 0.14 pJ/b inductive-coupling transceiver with digitally-controlled precise pulse shaping

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Miura, Noriyuki; Ishikuro, Hirok; Niitsu, Kiichi; Sakurai, Takayasu; Kuroda, Tadahiro

被引用次数: 29

2008

Closed-form expressions for crosstalk noise and worst-case delay on capacitively coupled distributed RC lines

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Kawaguchi, Hiroshi; Antono, Danardono Dwi; Sakurai, Takayasu

被引用次数: 0

2007

A large-area wireless power-transmission sheet using printed organic transistors and plastic MEMS switches

NATURE MATERIALS

Sekitani, Tsuyoshi; Takamiya, Makoto; Noguchi, Yoshiaki; Nakano, Shintaro; Kato, Yusaku; Sakurai, Takayasu; Someya, Takao

被引用次数: 338

2007

Conformable, flexible, large-area networks of pressure and thermal sensors with organic transistor active matrixes

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA

Someya, T; Kato, Y; Sekitani, T; Iba, S; Noguchi, Y; Murase, Y; Kawaguchi, H; Sakurai, T

被引用次数: 1424

2005

A large-area, flexible pressure sensor matrix with organic field-effect transistors for artificial skin applications

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA

Someya, T; Sekitani, T; Iba, S; Kato, Y; Kawaguchi, H; Sakurai, T

被引用次数: 1856

2004

A reduced clock-swing flip-flop (RCSFF) for 63% power reduction

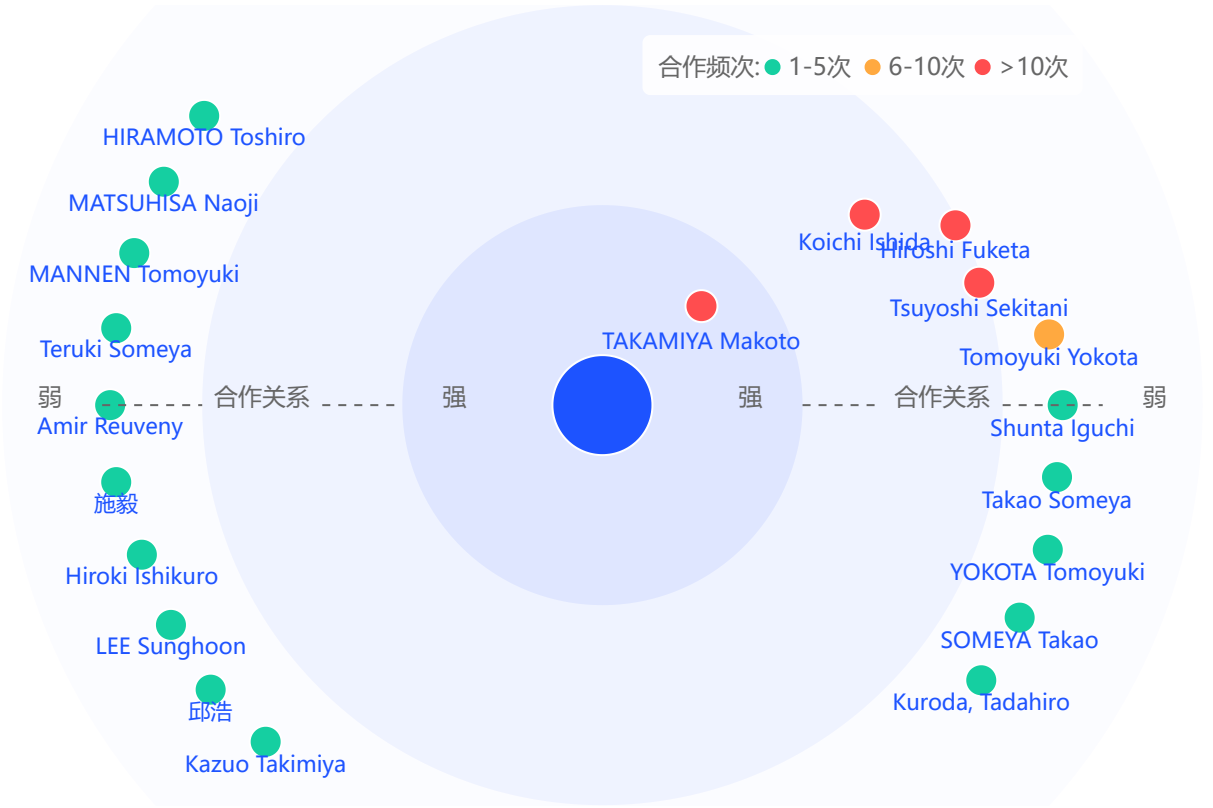
IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kawaguchi, H; Sakurai, T

被引用次数: 177

1998

学术合作



TAKAMIYA Makoto	合作次数: 45
东京大学	
Koichi Ishida	合作次数: 22
德累斯顿工业大学	
Hiroshi Fuketa	合作次数: 13
东京大学	
Tsuyoshi Sekitani	合作次数: 13
大阪大学	
Tomoyuki Yokota	合作次数: 6
东京大学	

还有 19 个合作者